WEAVE: Overcoming the Identity Shortcut in Style Transfer via 3-Minute Wavelet-Decoupled Flow Matching

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Abstract

Current style transfer methods face two primary challenges. First, state-of-the-art diffusion models, along with training-free methods built on them, require more than 60 GB of VRAM and therefore cannot run on consumer GPUs. Second, lightweight methods without these generative backbones often underperform a trivial identity mapping, which scores high on traditional metrics like ArtFID and LPIPS but was rarely evaluated as a baseline.

We empirically observe that frequency dominance in shared representation spaces pushes lightweight models toward identity-like solutions (Rahaman et al. 2019): low-frequency structural energy overwhelms high-frequency style gradients, a pattern we confirm through gradient-spectrum analysis and per-subband style-transfer measurements.

We argue that an effective style transfer method should outperform the identity baseline while remaining computationally affordable. To this end, we propose WEAVE: a single-level Haar wavelet transform splits the VAE latent into a low-frequency structure band and high-frequency style bands; a compact rectified-flow network learns a structure-preserving velocity field; and endpoint statistical alignment (AdaIN) injects the target style into the high-frequency subbands. This decoupling separates content transport from style injection, letting WEAVE train in 3 minutes on a single consumer GPU.

We also advocate the IDT–TGT sandwich as an intuitive evaluation guideline. IDT returns the source unchanged, while TGT returns an unrelated image from the target style; together they show whether an output gains style without giving up source content beyond the target-reference level. Under this guideline, WEAVE clears the identity reference with only 903K trainable parameters (plus a frozen 84M VAE), making high-quality style transfer both effective and affordable.

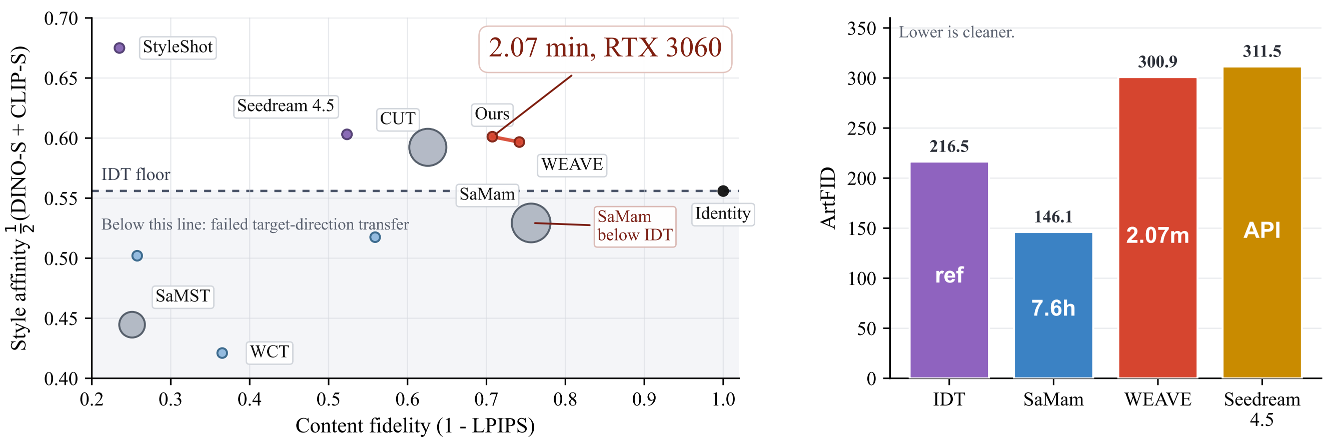

Figure 1: IDT–TGT sandwich comparison on Distinct5-WikiArt. Left: style affinity $\frac{1}{2}(\text{DINO-S} + \text{CLIP-S})$ versus $1 - \text{LPIPS}$, with bubble area encoding training time. The red points mark two WEAVE operating points. Square markers denote training-free diffusion editors (StyleAligned). The dashed line marks the IDT style reference (0.556); methods below it have not moved in the requested direction. The dotted line marks the TGT content reference (0.267); methods right of it are perceptually farther from the source than an unrelated target-style image. Right: target-pooled ArtFID on the same benchmark, with numbers inside the bars indicating training or response time.

Introduction

Style transfer faces two linked problems in deployment and evaluation. Large diffusion editors and training-free methods built on them can require more than 60 GB of GPU memory, which places their strongest priors outside consumer hardware.

Lightweight alternatives avoid this cost, but some preserve the source so strongly that their style affinity falls below the unchanged input. Conventional protocols can hide this no-op behavior: LPIPS and ArtFID strongly reward structural reconstruction, yet the identity mapping is rarely reported as an explicit baseline.

Figure 1 makes the missing comparison visible with two simple outputs. Identity (IDT) returns the source unchanged, while Target Style (TGT) returns an unrelated reference from the requested style. A useful transfer should move beyond IDT in the style direction without moving as far from the source as TGT in the content direction. The two references expose opposite failures in the original metric units: methods below IDT have not acquired enough target style, whereas methods beyond TGT have discarded at least as much source content as a source-independent style image. We call this the IDT–TGT sandwich. It is an empirical diagnostic, not a replacement for DINO, CLIP, LPIPS, or visual inspection.

Why do compact models enter the identity regime? We argue that an important driver is the shared coordinate system rather than parameter count alone. Neural networks learn low frequencies first (Rahaman et al. 2019; Xu, Zhang, and Luo 2022), and natural images place much more energy in global layout and object shape than in brushstroke and fine texture. A flow-matching model trained on the full latent must explain both with one velocity field. The resulting low-frequency residuals are larger and easier to fit, so their gradients dominate the update and weaken style-conditioned high-frequency motion. Frequency-domain translation reports a related content–style conflict (Cai et al. 2021); our gradient and subband probes measure it directly in the trained model. Simply increasing width does not change this competition because both signals still share the same basis and objective.

We therefore propose WEAVE, which changes the representation and target before increasing capacity. A frozen diffusion VAE supplies a compact latent, while a one-level Haar transform separates low-frequency structure (LL) from oriented texture (LH/HL/HH). A 903K-parameter rectified-flow model learns source-aligned transport with weaker LL supervision, and one endpoint AdaIN update supplies target-style statistics to the high-frequency bands. Cached latents remove the VAE encoder and diffusion U-Net from training; eight compact Euler evaluations and one VAE decode are sufficient at inference. WEAVE consequently trains in 3 minutes on one RTX 3060, and the same endpoint operator can be added to frozen SD1.5 without retraining (Section).

Contributions. (1) **Evaluation:** We advocate the IDT–TGT sandwich, which reports a no-op reference and a source-independent target-style reference alongside standard metrics, exposing both style suppression and content collapse. (2) **Representation and method:** We diagnose low-frequency gradient dominance in a shared latent objective and develop a wavelet-decoupled flow that assigns structure, learned texture transport, and endpoint statistics to explicit paths. (3) **Efficiency and evidence:** WEAVE uses 903K trainable parameters and 3 minutes of single-GPU training. Cross-dataset evaluation, gradient/subband interventions, and a frozen SD1.5 plugin test generalization and portability.

Related Work

Neural Style Transfer

Artistic style transfer was introduced by optimizing pixels to match content features and Gram statistics of a pretrained network (Gatys, Ecker, and Bethge 2015, 2016). AdaIN aligns channel-wise moments (Huang and Belongie 2017), while WCT aligns a full feature covariance (Li et al. 2017); both replace iterative optimization with a feed-forward feature transform. SANet and transformer-based variants add correspondence or long-range attention (Park and Lee 2019; Deng et al. 2022; Hong et al. 2023; Kwon et al. 2024), and CUT learns unpaired translation through patch-level contrast (Park et al. 2020). Training-free methods also reuse pretrained codebooks or hierarchical vision transformers (Gim, Yoo, and Uh 2024; Zhang et al. 2024a). The survey of Zhang et al. (Zhang and Tang 2025) summarizes this progression. These methods differ in supervision and backbone, but most still mix structure and appearance in one feature tensor. Stronger moment or attention transfer can therefore alter source geometry; WEAVE instead restricts the analytic style correction to explicit frequency coordinates.

Diffusion-based and Training-free Style Editing

Latent diffusion models (Rombach et al. 2022) move editing into a compressed generative space and provide strong semantic priors. Prompt-to-Prompt and SDEdit control generation through attention or noising (Hertz et al. 2022; Meng et al. 2021); StyleID and StyleAligned modify attention statistics (Chung, Hyun, and Heo 2024; Hertz et al. 2024); IP-Adapter adds an image prompt (Ye et al. 2023); and InST, DiffuseIT, ArtBank, and Z-STAR use inversion or pretrained features for style transfer (Zhang et al. 2023; Kim, Kwon, and Ye 2023; Zhang et al. 2024b; Lu et al. 2024). These methods are often called training-free because they do not update backbone weights, but inference still invokes a large denoiser repeatedly and may require inversion. Recent FLUX and Qwen-Image backbones further increase the memory cost (Black Forest Labs 2024; Alibaba Cloud 2024). WEAVE retains only the frozen VAE and learns the remaining transport with a compact model; the SD1.5 plugin separately tests whether its analytic endpoint operation transfers back to a large pretrained editor.

Flow Matching and Schrödinger Bridges

Rectified flow (Liu, Gong, and Liu 2023) and flow matching (Lipman et al. 2022) regress a velocity along an interpolation between endpoint distributions, enabling short numerical paths. Schrödinger bridges (Léonard 2014) add entropy-regularized stochastic transport, and DSB, DSBM, I2SB, and DDIB develop practical diffusion or flow solvers for unpaired domains (De Bortoli et al. 2021; Shi et al. 2023; Liu et al. 2023; Su et al. 2023). These methods are related to regularized optimal transport (Cuturi 2013; Peyré and Cuturi 2019), but their image objectives usually measure all coordinates together. WEAVE does not replace the rectified-flow path. It constructs a source-aligned endpoint, applies the same path in an orthogonal wavelet basis, and weights its blocks separately. This preserves the few-step advantage while changing which errors compete during learning.

Lightweight Models and Evaluation

Mamba-based SaMST and SaMam reduce trainable size and sequence cost (Liu et al. 2024, 2025; Gu and Dao 2023), showing that style transfer need not inherit a full diffusion U-Net. Efficiency alone, however, does not establish that the output moved toward the requested style. LPIPS measures perceptual source distance (Zhang et al. 2018), and ArtFID combines style distribution and content terms (Wright and Ommer 2022); both can reward a near-copy when the identity output is omitted. IDT makes this no-op score explicit. TGT adds the opposite reference: a target-style image with no source dependence. Reporting both rows preserves the standard metrics and their raw units, while identifying whether an apparent gain comes from style suppression or content collapse. The references are benchmark-specific diagnostics rather than universal thresholds.

Spectral Bias and Frequency-aware Optimization

Neural networks preferentially fit low-frequency functions (Rahaman et al. 2019; Xu, Zhang, and Luo 2022), and natural images place most spectral energy in coarse structure. The two effects are distinct but reinforce each other: the training target supplies larger low-frequency residuals, while the network fits those residuals earlier. Frequency-domain translation therefore constrains low and high frequencies separately to protect content identity (Cai et al. 2021). Our contribution is to connect this optimization imbalance to the identity shortcut in a compact flow model and measure the connection through per-band loss, input gradients, head gradients, and endpoint interventions. The measurements motivate a structural preconditioner rather than another scalar style weight.

Wavelet and Latent-space Alignment

Wavelets separate local information by scale and orientation and support exact reconstruction. They have been used in restoration and generation (Phung, Dao, and Tran 2023) and in photorealistic style transfer (Yoo et al. 2019). The Haar basis is particularly simple: its orthogonality gives additive energy decomposition, and its compact support preserves local correspondence (Daubechies 1992). VAE latents provide a complementary reduction in spatial cost (Rombach et al. 2022), while AdaIN and WCT provide closed-form statistical alignment. WEAVE combines these ingredients inside the transport objective rather than using wavelets only as preprocessing or postprocessing. LL, LH, and HL receive explicit velocity roles; HH is reserved for endpoint statistics. This division is the central difference from a single latent-space transform such as Latent-WCT.

Method

A direct content-to-style target changes layout, palette, and texture at once. Under shared latent MSE, the dominant structural residual sets the update, so increasing style strength often spends content instead of resolving the conflict. WEAVE instead constructs a source-aligned endpoint, learns transport in orthogonal Haar bands, and applies one high-frequency statistics update after the path (Figure 2).

Why a Shared Latent Objective Favors Structure

Consider a velocity error $e_\theta = v_\theta(z_t, s) - u$. Because the Haar transform is orthonormal, an unweighted latent MSE can be written exactly as

$$\mathcal{L}_{\text{shared}} = \mathbb{E}_t \|e_\theta\|_2^2 = \mathbb{E}_t \left[\|e_\ell\|_2^2 + \sum_{i=1}^{3} \|e_{h_i}\|_2^2 \right]. \quad (1)$$

The sum alone is not harmful; the problem is its scale and shared parameterization. Let $J_b = \partial v_b / \partial \theta$ be the output Jacobian of band b. The parameter update is

$$\nabla_\theta \mathcal{L}_{\text{shared}} = 2\,\mathbb{E}_t \left[J_\ell^\top e_\ell + \sum_{i=1}^{3} J_{h_i}^\top e_{h_i} \right]. \quad (2)$$

Natural-image geometry makes $\|e_\ell\|$ large, while spectral bias makes low-frequency functions easier to fit (Rahaman et al. 2019; Xu, Zhang, and Luo 2022). Unless the high-frequency Jacobians compensate, the LL term sets the early descent direction and the model underuses style conditioning. The measured parameter gradients show the same imbalance (Section). WEAVE changes the endpoint residual, separates output heads, and downweights LL before gradients enter the shared backbone.

Framework Setup

Let $z_c = \mathcal{E}(x_c) \in \mathbb{R}^{4\times32\times32}$ and $z_s = \mathcal{E}(x_s)$ be content and target-style latents from the frozen SD1.5 VAE (Rombach et al. 2022). A direct path from z_c to z_s also learns the unrelated geometry of x_s. We instead decompose $\mathcal{W}(z_c) = (\ell_c, h_{1,c}, h_{2,c}, h_{3,c})$ and similarly for z_s, then form the structure-aligned endpoint

$$\begin{aligned} \tilde{\ell}_s &= (1-\alpha)\ell_c + \alpha\,\mathrm{AdaIN}(\ell_c, \ell_s), \qquad \alpha = 0.3, \\ z_1^\star &= \mathcal{W}^{-1}(\tilde{\ell}_s, h_{1,s}, h_{2,s}, h_{3,s}). \end{aligned} \quad (3)$$

The LL blend permits coarse appearance to move while keeping source layout; the high-frequency target comes from the requested style. Exact source LL suppresses palette change, whereas the complete style latent would also supervise unrelated target geometry.

The corresponding target velocity is explicit in wavelet coordinates. Writing $\mathcal{W}(z_1^\star - z_c) = (u_\ell, u_{h_1}, u_{h_2}, u_{h_3})$,

$$\begin{aligned} u_\ell &= \alpha[\mathrm{AdaIN}(\ell_c, \ell_s) - \ell_c], \\ u_{h_i} &= h_{i,s} - h_{i,c}, \qquad i \in \{1,2,3\}. \end{aligned} \quad (4)$$

Thus α bounds the supervised coarse movement, while the high-frequency residuals retain the target-style signal. This is different from merely reducing the LL loss weight: Eq. 4 changes what the model is asked to predict, whereas the weight controls how strongly an error changes the parameters.

Rectified flow learns the straight path

$$z_t = (1-t)z_c + t z_1^\star, \qquad u = z_1^\star - z_c, qquad t \sim \mathcal{U}([0,1]). \quad (5)$$

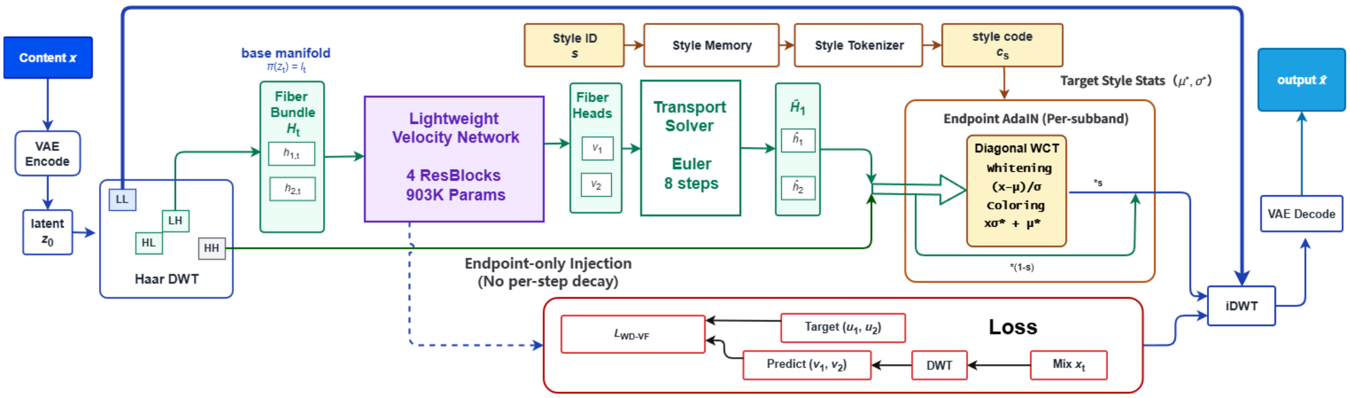

Figure 2: **WEAVE architecture**. A single-level Haar transform exposes the LL/LH/HL/HH subbands of the SD1.5 VAE latent. The compact velocity network v_θ ($\sim$903K) predicts motion for LL/LH/HL via three heads, while HH carries no head. Band-weighted rectified-flow matching learns the transport; per-subband AdaIN aligns LH/HL/HH once at the endpoint with scales $0.3/0.3/0.5$ and leaves LL unchanged.

Because $\mathcal{W}$ is linear, the same interpolation holds independently in every subband. WEAVE therefore keeps the standard rectified-flow target and changes only the endpoint, parameter sharing, and loss weights.

Wavelet Decomposition and Frequency Routing

Low-frequency structure carries much more energy than fine texture, so a shared loss favors source reconstruction (Rahaman et al. 2019; Xu, Zhang, and Luo 2022; Cai et al. 2021). WEAVE separates these signals before predicting the flow.

Haar wavelet decomposition. A one-level Haar transform gives $\mathcal{W}(z_t) = (\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$: one LL structure band and three high-frequency bands, each $4\times16\times16$. Haar is orthogonal and local, so high-frequency updates do not directly alter the LL coordinate (Daubechies 1992).

$$\|\Delta z\|_F^2 = \|\Delta \ell\|_F^2 + \sum_{i=1}^{3} \|\Delta h_i\|_F^2. \tag{6}$$

Equation 6 is simply Parseval's identity. It makes the design transparent: an operation restricted to LH/HL/HH leaves the LL coefficient unchanged, although the nonlinear VAE decoder can still couple visible color and texture.

Separate 1×1 heads then remove output-level interference even though the backbone is shared: the aligned endpoint defines the movement, λ_{LL} controls structural gradient pressure, and AdaIN handles statistics that spatial MSE learns poorly.

Empirical frequency probes. Across 5,000 cross-style latent pairs, transport energy concentrates in LL while style separation is strongest in HH (Figure 3). This motivates a weak LL loss and stronger high-frequency routing.

Architecture. The 903K-parameter velocity network has four width-64 residual blocks and separate 1×1 heads for LL, LH, and HL. HH has no velocity head and changes through the endpoint operation. Learned style-memory tokens provide a category-level prior to the shared backbone; the ablations show

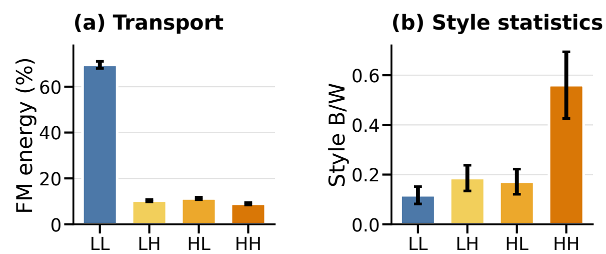

Figure 3: **Frequency probes.** (a) Cross-style transport energy in each Haar band. (b) Between-style versus within-style separation of latent statistics. Error bars summarize variation across style routes or statistics.

that this branch is auxiliary rather than the main style-injection path. HH is deliberately statistical rather than transported. Although it carries stroke and grain statistics, its phase is not registered in unpaired transfer: $h_{3,s}(i,j)$ is not the desired detail for source location (i,j). A pointwise HH head would minimize $\|v_{h_3} - (h_{3,s} - h_{3,c})\|_2^2$, whose optimum averages incompatible phases and sends noisy gradients through the shared backbone. Endpoint AdaIN instead matches per-sample HH moments without learning target-image coordinates; a matched HH-head ablation did not improve the style-content frontier.

Why the model is fast. For a 512×512 image, the frozen VAE produces only 32×32 spatial positions; the velocity backbone operates on 16×16 subbands, versus 512×512 RGB positions. Cached latents remove the encoder and the large diffusion U-Net from training. Rectified flow supplies target velocities directly, so there is no adversarial loop, denoising teacher, or inversion stage; inference uses a short path, one analytic transform, and one VAE decode.

With C_v for one velocity evaluation and C_{dec} for one frozen VAE decode,

$$C_{\text{WEAVE}} \approx K C_v + C_{\text{AdaIN}} + C_{dec}. \tag{7}$$

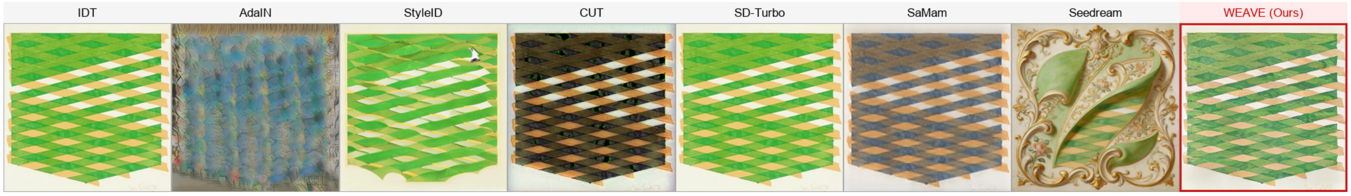

Figure 4: Extreme cross-domain stylization (Minimalism $\to$ Rococo). Diffusion editors substantially alter source structure, while several lightweight or classical outputs remain close to identity. WEAVE preserves source topology through low-frequency anchoring and injects painterly texture through high-frequency subbands.

Here C_v acts on small subbands and C_{AdaIN} is a linear-time moment update, whereas training-free diffusion editors still repeatedly evaluate a large denoiser.

Spectral flow-matching objective. The spectral flow-matching term is

$$\mathcal{L}_{\mathrm{SFM}} := \mathbb{E}_t\big[\lambda_{LL}\|v_\ell - u_\ell\|_2^2 + \|v_{h_1} - u_{h_1}\|_2^2 + \|v_{h_2} - u_{h_2}\|_2^2\big], \tag{8}$$

with $\lambda_{LL} = 0.3$. The lower LL weight limits structural dominance while still allowing coarse tone to move. The predicted endpoint is $\hat{z}_1 = z_c + \mathcal{W}^{-1}(v_\ell, v_{h_1}, v_{h_2}, 0)$.

Equation 8 is a diagonal preconditioner. With $D = \mathrm{diag}(\lambda_{LL} I, I, I, 0)$ and $e = \mathcal{W}(v - u)$, its output-space gradient is $2De$. The relative LL pressure before the shared Jacobian is therefore

$$r_{\mathrm{out}} = \frac{\lambda_{LL}\|e_\ell\|_2}{(\|e_{h_1}\|_2^2 + \|e_{h_2}\|_2^2)^{1/2}}. \tag{9}$$

Choosing $\lambda_{LL} = 0.3$ reduces this pressure without deleting the coarse target. Parameter gradients are not perfectly separated because the backbone is shared, which is why Section measures them directly. HH is omitted from the velocity loss because the network has no HH head; its style statistics are supplied at the endpoint.

Endpoint AdaIN: Primary Style Injection Channel

Flow matching alone carries a weak style signal. WEAVE therefore applies AdaIN once, after the last Euler step, to align high-frequency statistics without repeatedly disturbing the transport path. Removing this endpoint update largely returns the model to source preservation (Table 2).

Alignment operator. Let $\hat{h}_i$ and $h_i^\star$ be high-frequency subbands of the predicted latent and a target-style reference. For $i \in \{1,2,3\}$, AdaIN (Huang and Belongie 2017) gives

$$T_i(a) = \frac{a - \hat{\mu}_i}{\hat{\sigma}_i}\,\sigma_i^\star + \mu_i^\star, \tag{10}$$

where μ and σ are channel-wise spatial statistics. We blend this update with scales $(s_1, s_2, s_3) = (0.3, 0.3, 0.5)$:

$$\hat{h}_i^+ = (1 - s_i)\hat{h}_i + s_i\,T_i(\hat{h}_i), \qquad i \in \{1,2,3\}, \tag{11}$$

LL is left unchanged in this endpoint operation.

For positive channel standard deviations, the update has a direct statistical meaning:

$$\begin{aligned}\mu(\hat{h}_i^+) &= (1 - s_i)\hat{\mu}_i + s_i\mu_i^\star,\\ \sigma(\hat{h}_i^+) &= (1 - s_i)\hat{\sigma}_i + s_i\sigma_i^\star.\end{aligned} \tag{12}$$

The scale s_i therefore specifies how far each moment moves toward the reference, without requiring the velocity network to infer this global correction from spatial MSE. Applying the map only after the final Euler step keeps the correction out of later velocity updates. LL anchoring protects layout, the learned LH/HL path carries organized texture, and Table 1 shows that endpoint statistics alone are insufficient.

Training and Inference

Training adds a light edge ℓ_1 term and low-pass anchor to Eq. 8; together they contribute less than 5% of measured gradient mass. At inference, $z^0 = z_c$ and, for $K = 8$,

$$z^{k+1} = z^k + \frac{1}{K}\,\mathcal{W}^{-1}\big(v_\ell^k, v_{h_1}^k, v_{h_2}^k, 0\big), \tag{13}$$

then apply high-frequency AdaIN once and decode with the frozen VAE. Full architecture, training fields, and step ablations are in the supplement.

Experiments

Setup and Protocol

Datasets. Distinct5-WikiArt-512 (D5) contains five deliberately separated styles, with 3,600 training and 150 test images per style; its $5{\times}5{\times}30$ board has 750 requests. P2A-256 tests lower resolution, and R5 expands to all 20 WikiArt families (12,000 requests). DINO-S, our primary style metric, compares output DINOv2 features with the target-style pool; CLIP-S is secondary. DINO-C and LPIPS measure source retention. Table 1 compares twelve representative baselines under all four views.

IDT copies the source and TGT returns a fixed source-independent target-style image. Table symbols mark values outside the resulting bracket for that board and metric, not global method rankings. Validation uses 50 held-out pairs; boards, reference pools, preprocessing, and seeds are fixed. Full construction and hyperparameters are in the supplement.

Main Results

Reading the sandwich. The marks expose two operating regimes without changing the metrics: some rows remain

Method	D5-512				P2A-256				R5-WikiArt				Params	Train	Infer
	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑		min	750 imgs
Identity	0.419	0.693	0.000	1.000	0.416	0.663	0.000	1.000	0.433	0.731	0.000	1.000	0	free	0 s
Target Style	1.000	0.874	0.733	0.176	1.000	0.835	0.800	0.169	1.000	0.796	0.747	0.230	0	free	0 s
SD-Turbo	0.484	0.693†	0.003	0.922	0.341†	0.674	0.603	0.279	0.505	0.767	0.449	0.538	$0^{+1.1B}$	free	5 m
StyleAligned	0.675	0.780	0.869‡	0.239	0.612	0.768	0.786	0.310	0.649	0.824	0.829‡	0.315	$0^{+1.1B}$	free	77 m
Z-STAR	0.449	0.784	0.347	0.549	0.514	0.786	0.332	0.526	0.514	0.822	0.384	0.526	$0^{+1.1B}$	free	3 h
StyleShot	0.563	0.787	0.765‡	0.377	0.552	0.774	0.713	0.335	0.484	0.787	0.795‡	0.380	$0^{+2.4B}$	free	5.1 h
StyleID	0.548	0.822	0.552	0.416	0.256†	0.648†	0.691	0.147‡	0.571	0.815	0.493	0.374	$0^{+1.1B}$	free	63 m
CUT	0.471	0.714	0.374	0.795	0.539	0.754	0.494	0.702	0.441	0.710†	0.620	0.795	7.0M	322.6	5 m
SaMST	0.271†	0.618†	0.749‡	0.145‡	0.501	0.709	0.398	0.838	0.461	0.667†	0.612	0.615	8.3M	39.5	10 m
SaMam	0.477	0.582†	<u>0.243</u>	0.812	0.505	0.677	<u>0.205</u>	0.870	0.503	0.712†	<u>0.227</u>	0.925	8.8M	436.0	17.6 m
StyTR-2	0.501	0.709	0.596	0.747	0.515	0.722	0.495	0.555	0.533	0.759	0.553	0.737	$35.4M^{+12.9M}$	–	4 m
AesPA-Net	0.520	0.724	0.548	0.601	0.498	0.744	0.571	0.522	0.545	0.782	0.409	0.728	$11.3M^{+20.0M}$	–	3 m
Seedream 4.5	0.486	0.720	0.477	0.739	0.517	0.752	0.227	0.785	0.504	0.742	0.486	0.725	–	API	–
Latent-WCT	0.362†	0.673†	0.441	0.559	0.338†	0.738	0.585	0.386	0.414†	0.710†	0.444	0.553	0	free	18 s
Ours	**0.4859**	0.7075	**0.2583**	**0.8287**	0.4801	0.6681	0.3116	0.8612	<u>0.5226</u>	0.7747	0.2895	0.7717	$\mathbf{903K}^{+84M}$	**3.0**	**95 s**

Table 1: Main comparison. DINO-S, CLIP-S, and DINO-C are higher better; LPIPS is lower better. IDT and TGT are empirical references, not theoretical bounds. †: style value does not exceed IDT. ‡: content value does not remain inside TGT (LPIPS $\geq$ TGT or DINO-C $\leq$ TGT). These marks diagnose this board and metric only. Superscripts in Params denote frozen parameters. Train is wall time on one RTX 3060; "free" marks training-free methods, "–" marks author-released pretrained weights used without retraining, and "API" marks closed models. Infer is generation-only time for 750 images on one RTX 3060.

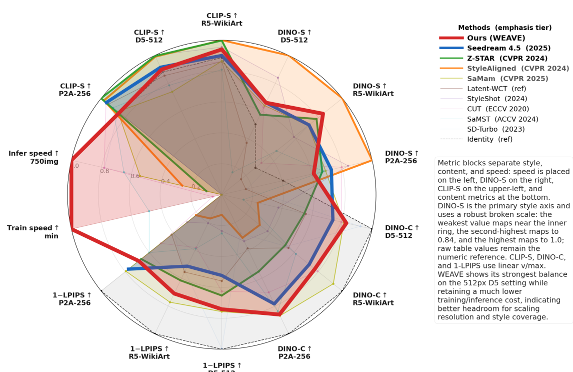

Figure 5: Style, content, and cost across three benchmarks. DINO-S uses a broken visual scale to separate the middle tier; other axes are linearly normalized. Table 1 gives raw values.

below IDT on style, while others cross TGT on content distance. Unmarked rows show that the bracket is attainable by very different architectures. Its board dependence also motivates computing IDT and TGT separately rather than using one absolute threshold.

Cross-metric interpretation. The sandwich is intentionally metric-specific. DINO-S and CLIP-S use different visual representations and need not rank texture, palette, and semantics in the same order. Likewise, LPIPS emphasizes local perceptual change whereas DINO-C measures semantic feature retention. Agreement across the two style and two content views is therefore stronger evidence than any single score; disagreement identifies which aspect of the transfer moved. This is why we retain raw metrics rather than collapsing the bracket into a composite index.

D5 result. WEAVE clears IDT on both style measures while retaining substantially more source content than editors with a similar or stronger style score. Among trained methods that clear the CLIP-S identity reference, it has the lowest LPIPS and is the only sub-million-parameter network. The comparison with Seedream is especially informative: similar DINO-S does not imply a similar content cost.

P2A and R5. WEAVE remains inside the bracket on P2A, although SaMam gives the best measured trade-off there. On R5, WEAVE reaches the strongest DINO-S among the trained lightweight methods while retaining a low source distance as the style space grows. The stronger 512-resolution profile is consistent with the wavelet route benefiting from high-frequency detail that is compressed at 256, suggesting favorable scale potential. Because P2A and D5 differ in data as well as resolution, we treat this as evidence for scaling rather than an isolated resolution ablation. Figure 5 summarizes the full profile.

Across the three boards, the same compact path remains content-preserving as resolution and style coverage increase. The stronger 512 profile matches the method's mechanism: native high-frequency coefficients preserve brushstroke and edge statistics that are compressed at lower resolution, while the LL anchor does not grow a new semantic burden. This suggests that additional resolution can improve style capacity without requiring a proportionally larger transport network.

Cost. WEAVE is substantially cheaper to train and run than the learned baselines in Table 1. Latent-WCT is faster still but falls below IDT, showing that analytic wavelet statistics alone do not replace learned transport.

Mechanistic Ablations

Table 2 compares matched diagnostic variants on D5. Removing AdaIN or flow matching shifts the result toward source preservation, while removing cross-attention changes every metric by at most 0.010. One Euler step and strong extrapolation degrade both objectives, identifying the active paths and their stable operating range.

Config	CLIP-S↑	LPIPS↓	DINO-S↑	DINO-C↑
Baseline	**0.727**	**0.343**	**0.483**	**0.755**
w/o AdaIN	0.708	0.334	0.484	0.816
w/o Flow Matching	0.710	0.296	0.475	0.819
w/o Cross-attention	0.726	0.336	0.486	0.765
λ_{LL}=0	0.713	0.313	0.480	0.804
lr = 5×10^{-4}	0.730	0.356	0.487	0.738
AdaIN α=0	0.709	0.339	0.483	0.811
AdaIN α=2.0	0.717	0.297	0.485	0.802
extrap = 1.0	0.704	0.591	0.411	0.616
num_steps = 1	0.709	0.476	0.376	0.448

Table 2: Ablation on D5 (750 pairs). The reported main model uses 10 epochs and a 0.3 LL target blend. Changes below 0.002 CLIP-S are omitted.

Setting $\lambda_{LL} = 0$ moves the model toward the source, so weak LL supervision is useful for coarse tone. Additional diagnostic sweeps are reported in the supplement.

Generalization and Robustness

Representation and adaptation. A matched RGB control performs worse on both style and content. Updating only the style memory from 30 images clears IDT on eight held-out styles, with little gain from full fine-tuning.

Unseen families. On five unseen WikiArt families, WEAVE gives the highest CLIP-S and DINO-C and the lowest LPIPS (Table 3); SaMam is 0.015 higher on DINO-S but has higher LPIPS.

Method	CLIP-S ↑	LPIPS ↓	DINO-S ↑	DINO-C ↑
Ours (WEAVE)	**0.7742**	**0.2802**	0.5313	**0.7587**
SaMam	0.7577	0.3990	**0.5466**	0.7369
SaMST	0.7485	0.6213	0.3913	0.2680

Table 3: Zero-shot evaluation on Other5 (750 pairs) using the D5 checkpoint.

Basis and resolution. Daubechies-2 slightly improves CLIP-S but worsens LPIPS; Haar gives the better measured trade-off and has exact orthogonality with compact support. Native-resolution training also outperforms upsampling a low-resolution checkpoint, consistent with style information being carried by high-frequency bands that low-resolution training compresses.

Plug-and-Play Extension to Pre-trained Diffusion Models

We add the same Haar and endpoint AdaIN operation to a frozen SD1.5 image-to-image pipeline without training. The base editor uses an empty prompt, strength 0.5, and 20 DDIM steps; the plugin aligns LH/HL/HH at scale 0.8 while leaving LL unchanged. On 600 off-diagonal D5 pairs, both CLIP-S and DINO-S increase ($p<10^{-15}$, paired t-test) with little content change, and the improvement remains on the full 750-pair board. The DWT–AdaIN–IDWT operation adds 4.9 ms per image. This experiment tests transferability of the operator, not a claim that the small network and SD1.5 are otherwise equivalent.

Mechanism Analysis

Gradient spectrum. At an intermediate flow time, the unweighted LL loss and its head gradient dominate the high-frequency terms. LL also remains the largest input-gradient band after down-weighting, so the imbalance is not created solely by the chosen loss weights. This supports frequency dominance in the trained system, although it does not prove that every identity-like solution has the same cause.

Subband roles. Output probes show that LH/HL carry most of the measured style change, LL retains structure, and headless HH changes only through endpoint AdaIN. Inference-time module swaps agree: removing AdaIN sharply reduces the LH style-transfer ratio, while replacing it with full-covariance WCT adds content drift for little style gain. Together with the ablation table, these probes support the intended division of labor between learned transport and endpoint alignment.

Discussion

WEAVE separates three responsibilities that compete in a shared latent loss: the aligned target limits coarse displacement, the flow learns spatially organized residual transport, and endpoint AdaIN supplies high-frequency moments. The evidence is consistent across levels. Training probes measure LL-dominant loss and gradients; component removal moves the result toward identity; and Latent-WCT shows that statistics without learned transport are insufficient. These results support frequency dominance as an important cause of the identity shortcut in this system, not as a universal explanation for every architecture.

Scope and limitations. IDT and TGT are empirical references rather than perceptual bounds; a flag motivates visual inspection, not a verdict. WEAVE is domain-conditional rather than reference-image conditional. D5 contains five separated styles, partly addressed by R5 and Other5, while LL protection makes palette-dominated or multi-scale styles harder. Human evaluation and more checkpoint families are needed to test the mechanism beyond this setting.

Conclusion

The IDT–TGT sandwich gives style and content metrics an explicit direction: IDT identifies methods stuck near the source, while TGT identifies those that abandon it. WEAVE addresses this conflict in frequency coordinates. A source-aligned target limits unnecessary structure motion, band-weighted flow learns organized transport, and endpoint AdaIN supplies high-frequency statistics without repeatedly perturbing the trajectory.

Experiments across datasets, unseen styles, and complementary metrics support this division. Gradient and subband probes link the identity shortcut to low-frequency dominance, while interventions show that analytic statistics and learned transport are both necessary. The native high-resolution and frozen-SD1.5 results suggest scale and transfer potential. FD-DB frames the same appearance–geometry tension in synthetic-to-real translation (Zang, Hu, and Song 2026), pointing to structure-preserving latent augmentation.

Ethics Statement

Style transfer tools can be misused to produce deceptive images or to misattribute stylistic authorship. Our method operates at the domain level (e.g., "Impressionism") rather than the per-artist level. This reduces the risk of imitating a specific living artist. We do not condition on individual reference paintings. The dataset is sourced from WikiArt and contains historically significant artworks. It does not contain personal data. We encourage downstream users to disclose when an image has been style-transferred and to respect source-content licensing terms.

Reproducibility Statement

All experiments use public data (WikiArt) and a public VAE (Stable Diffusion v1.5 EMA VAE). The experiments section gives the training configuration, hyperparameters, and evaluation protocol. Training takes 3 minutes on a single RTX 3060 (12 GB; 10-epoch wall-clock, batch 96). Generation-only inference for 750 pairs needs about 95 seconds (8-step ODE + VAE decode; $\approx$71 ms network generation and $\approx$52 ms decode per image), and a full metrics pass (CLIP/LPIPS on top of generation) is about 2 minutes. The 903K trainable parameter backbone checkpoint is 3.5 MB. We will release the code, the checkpoint, and the 750-pair evaluation protocol upon publication.

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